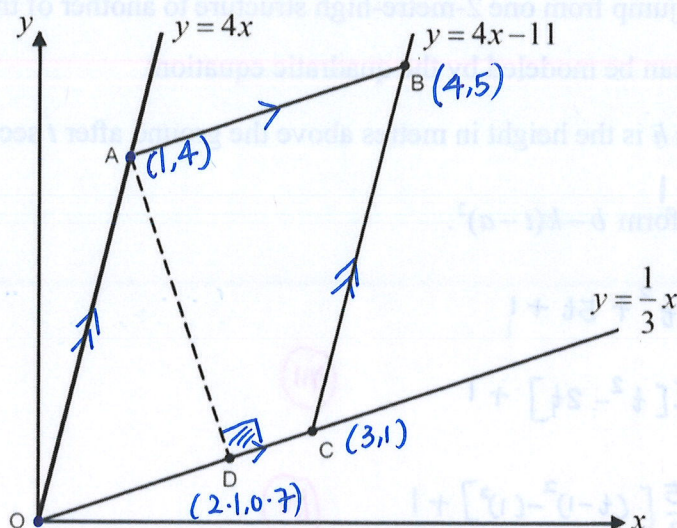


11



The diagram shows a parallelogram $OABC$ where O is the origin and A is the point $(1,4)$.

The equations OA , OC and CB are $y = 4x$, $y = \frac{1}{3}x$ and $y = 4x - 11$ respectively.

The perpendicular from A to OC meets OC at the point D .

Find

- (a) the coordinates of C , of B and of D .

[6]

$$\begin{aligned} C: \quad y &= 4x - 11 & \text{--- ①} & \text{Eqn of CB} \\ y &= \frac{1}{3}x & \text{--- ②} & \text{Eqn of OC} \end{aligned}$$

$$\begin{aligned} \text{①} = \text{②}, \quad 4x - 11 &= \frac{1}{3}x & \text{(m)} \\ \frac{11}{3}x &= 11 \end{aligned}$$

$$x = 3$$

$$y = \frac{1}{3}(3) = 1$$

$$\therefore C(3, 1) \quad \text{(A)}$$

$$D: \quad \text{Let } D(m, \frac{1}{3}m)$$

$$\text{grad}_{AD} = -3 \quad (\because AD \perp OC)$$

$$\frac{\frac{1}{3}m - 4}{m - 1} = -3 \quad \text{(m)}$$

$$\frac{1}{3}m - 4 = -3(m - 1)$$

$$\frac{1}{3}m - 4 = -3m + 3$$

$$\frac{10}{3}m = 7$$

$$\therefore m = 2.1, \quad \frac{1}{3}m = 0.7$$

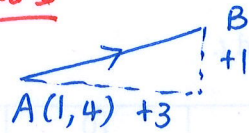
$$\therefore D(2.1, 0.7) \quad \text{(A)}$$

[Turn over]

(a) continuation . . .

(b) area of parallelogram OABC.

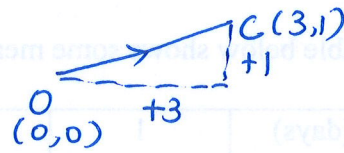
[2]

Method 1

$$\therefore B(4, 5)$$

(A1)

$$\begin{pmatrix} 1 \\ 4 \end{pmatrix} + \begin{pmatrix} 3 \\ 1 \end{pmatrix} = \begin{pmatrix} 4 \\ 5 \end{pmatrix}$$



$$O \rightarrow C$$

(M1)

$$\begin{pmatrix} 0 \\ 0 \end{pmatrix} + \begin{pmatrix} 3 \\ 1 \end{pmatrix} = \begin{pmatrix} 3 \\ 1 \end{pmatrix}$$

Let $B(h, k)$

Method 2 Since OABC is a //logram
OR Midpoint $A C = \text{Midpoint } O B$

$$\left(\frac{1+3}{2}, \frac{4+1}{2} \right) = \left(\frac{0+h}{2}, \frac{0+k}{2} \right)$$

$$\left(2, \frac{5}{2} \right) = \left(\frac{h}{2}, \frac{k}{2} \right)$$

$$\therefore h = 4$$

$$k = 5$$

$$\therefore B(4, 5)$$

(M1)

(b) Area of //logram OABC

$$= \frac{1}{2} \begin{vmatrix} 0 & C & B & A & O \\ 0 & 3 & 4 & 1 & 0 \end{vmatrix}$$

$$= \frac{1}{2} \left[(0+15+16+0) - (0+4+5+0) \right]$$

$$= \frac{1}{2} [31 - 9]$$

$$= \frac{1}{2} [22]$$

$$= 11 \text{ units}^2$$

(A1)

Method (3)

$$\text{Eqn } AB: y = \frac{1}{3}x + \frac{11}{3}$$

$$\text{Eqn } BC: y = 4x - 11$$

$$\vdots$$

$$B(4, 5)$$